

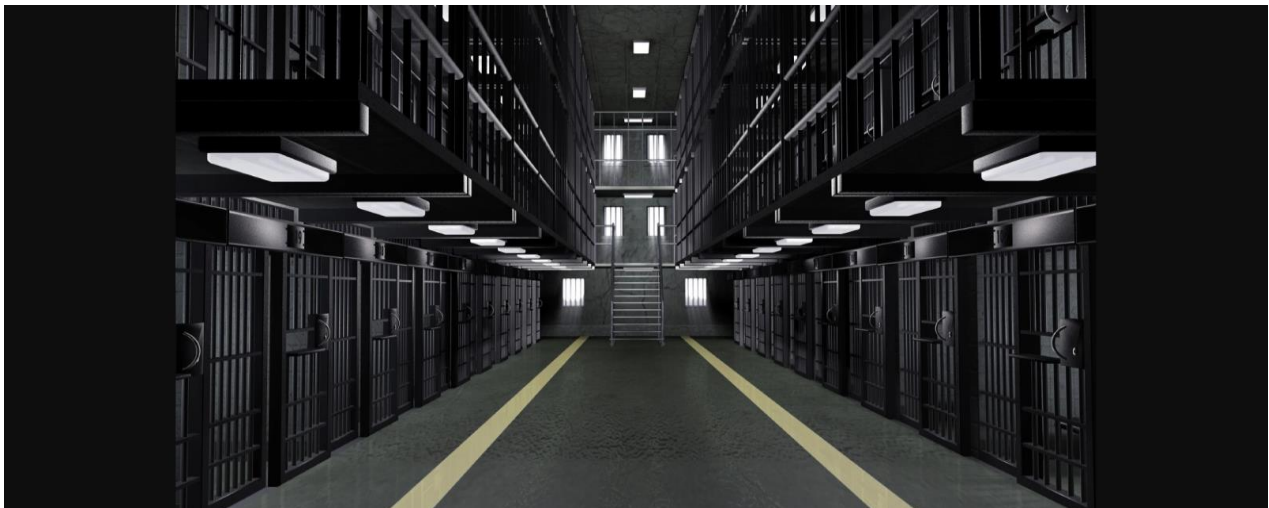
NATIONAL INSTITUTE OF BUSINESS MANAGEMENT

DIPLOMA IN SOFTWARE ENGINEERING(DSE23.1F)

COURSEWORK ONE

ELECTRONICS AND COMPUTER ARCHITECTURE

Prison System

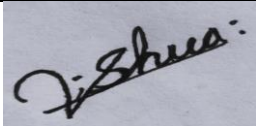
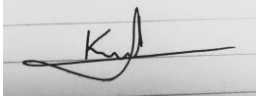
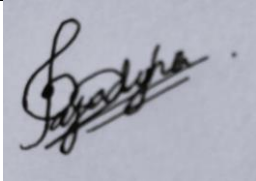
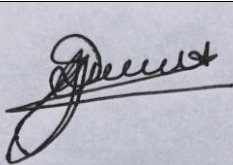


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DATE OF SUBMISSION: 8th of August 2023

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SUMMARY

- This system contains 5 main functions. They are,
 - **Identify and count the prisoners**
 - **Lighting System**
 - **Water Tank System**
 - **Security System**
 - **Vehicle Park System**
 - **High-Security System**
- We used LEDs, LDRs, Buzzers, and Motors to show the final output of the system. Also, we used Switches, Resistors, Transistors, and thermistors as the inputs. Using these electronic components our system will work according to the needed requirements.
- Using truth tables, k maps, and circuit diagrams these functions will be simplified and presented to the reader.

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CHAPTER 01

1.1 Introduction

What is a Prison System?

- ★ When a person was caught by doing illegal activities, they are sent to the jury. If the jury found the person guilty, they will be sent to a place where they were been locked up till their prison sentence is over. This place is called a **Prison**.
- ★ Inside a prison system, the security system is considered the most. Because there can be escape attempts and violence. So, many jailers and officers are there to check the prisoners. There are various kinds of security levels based on the type of prisoner.

1.2 Features of the Prison System

- **Maximum Security:** Since prison is a place where most dangerous criminals are looked it needs a good security system. So, there are many features like automatic-door locking systems, high-quality CCTV cameras, and many more.
- **Medical Services:** Sometimes prisoners can get into violent attacks and be gravely injured. So, every prison must have a good medical care field.
- **High-Security Systems:** When a vehicle comes and leaves the prison it needed to be checked carefully because people can bring illegal stuff like drugs, weapons, etc... So, prisons use features like laser scan systems to check the stuff people bring inside the prison.
- **Geographical location:** When the government constructs a prison, they will first check the locations where to build it. Because it needs to build somewhere isolated and be difficult to escape.

CHAPTER 02

2.1 Circuit Components

- **Breadboard**

It connects the power supplies (Arduino Uno Board) and the connecting wires which are required to develop a circuit.

- **Arduino Uno Board**

A normal battery supplies 1.5V. So, to get 5V to the circuit we can get 4 batteries and connect them. Then we get 6V to the circuit. Because of its high voltage circuit components can be burned. So, it is best to have an Arduino Uno Board, because it gives the exact voltage required (5V).

In the Arduino Uno board, GND will be connected to the breadboard's minus side and the 5V will be connected to the plus side.

- **Integrated Circuits**

Integrated circuits are made by using 14 pins. The 7th pin is used for GND, and the 14th pin is for VCC. Other pins will act as the inputs and outputs of the circuit.

According to the serial number in IC, the logic gate will be changed. Such as,

- AND Gate ic (7408)
- OR Gate ic (7432)
- NOT Gate ic (7404)
- NAND Gate ic (7400)

- **Connecting Wires**

To connect the components to make the circuit connecting wires are used.

- **LEDs**

To present the output of the circuit LEDs are used. There are assorted colours of LEDs such as green, red, yellow, etc...

- **LDR**

LDRs are used to measure the light intensity of a circuit. It also indicates when light is present.

- **Buzzers**

Buzzers are an electronic component that acts as an alarm.

- **Resistors**

Resistors are used to reduce the current flow. In it, there are several line strips. From those lines, the power of the resistor will be identified.

2.2 Truth Tables

- **Identify and count the prisoners**

- If the officer and the jailer both noticed that a prisoner did not arrive back from the working period, they both must press the button to warn others. Also, the cleaner is not allowed to turn on the switch.

A = Officer , B = Jailer, C = Cleaner

A	B	C	Z
0	0	0	0
0	0	1	0
0	1	0	1
0	1	1	0
1	0	0	1
1	0	1	0
1	1	0	1
1	1	1	0

- **Lighting System**

- Turn on the lights, jailers, and officers' use switches. They both can turn it on and off anytime. Also, it is security guard must turn on the lights at night.

A = Officer , B = Jailer, C = Security Guard

A	B	C	Z
0	0	0	0
0	0	1	1
0	1	0	1
0	1	1	1
1	0	0	1
1	0	1	1
1	1	0	1
1	1	1	1

- There are 3 buttons in the light system. 1st button presents the morning, 2nd button presents the daytime and the 3rd button presents the night-time. Only the lights will work if somebody presses the button 3.

Button1 = A, Button2 = B, Button3 = C

A	B	C	Z
0	0	0	0
0	0	1	1
0	1	0	0
0	1	1	1
1	0	0	0
1	0	1	1
1	1	0	0
1	1	1	0

- The prison has an emergency button to indicate if an emergency occurred. At least two people need to press the button, and if you did it will turn on the lights around the prison. (People; watcher in the watch tower, officer, and the jailer)

A = Jailer, B = Officer, C = Watcher

A	B	C	Z
0	0	0	0
0	0	1	0
0	1	0	0
0	1	1	1
1	0	0	0
1	0	1	0
1	1	0	1
1	1	1	1

• Water Tank System

- When a worker turned on the switch, the water tank will start to work. Also, the worker is only allowed to turn on the switch in the daytime.

A = Daytime, B = Switch

A	B	Z
0	0	0
0	1	0
1	0	0
1	1	1

• Security System

- When both the jailer and the officer turned on the emergency button, cells will be auto-locked, and a buzzer will work. They cannot activate it alone. But a watcher from a watch tower can.

A = Jailer, B = Officer, C = Watcher

A	B	C	Z
0	0	0	0
0	0	1	1
0	1	0	0
0	1	1	1
1	0	0	0
1	0	1	1
1	1	0	1
1	1	1	1

- If the smoke detector and the heat detector both start to work or, if a worker pressed the switch (fire alarm bell switch), the alarm and the fire exit door will be opened.

A = Smoke Detector, B = Heat Detector, C = Switch Alarm

A	B	C	Z
0	0	0	0
0	0	1	1
0	1	0	0
0	1	1	1
1	0	0	0
1	0	1	1
1	1	0	1
1	1	1	1

- Vehicle Park System**

- 3 people are allowed to open the parking gate. They are security guards, jailers, and the officers. The jailer is not allowed to open the gate by himself, he needs to do it with the officer. The security guard can do it by himself.

A = Security guard, B = Jailer, C = Officer

A	B	C	Z
0	0	0	0
0	0	1	1
0	1	0	0
0	1	1	1
1	0	0	1
1	0	1	1
1	1	0	1
1	1	1	1

- Add a password using 3 switches. If all the 3 switches were pressed, the parking gate will open. Also, the vehicle's number plate will be scanned too.

A = Switch1, B = Switch2, C = Switch3, D = Scanner

A	B	C	D	Z
0	0	0	0	0
0	0	0	1	0
0	0	1	0	0
0	0	1	1	0
0	1	0	0	0
0	1	0	1	0
0	1	1	0	0
0	1	1	1	0
1	0	0	0	0
1	0	0	1	0
1	0	1	0	0
1	0	1	1	0
1	1	0	0	0
1	1	0	1	0
1	1	1	0	0
1	1	1	1	0
1	1	1	0	0
1	1	1	1	1

- To turn on the lights there are 4 switches. If somebody pressed switch 1 and switch 2, all the lights will be turned on. Also, switch 3 and 4 will turn on all the lights. If you pressed all the switches, then also the lights will turn on. Else lights will not turn on.

Switch1 = A, Switch2 = B, Switch3 = C, Switch4 = D

A	B	C	D	Z
0	0	0	0	0
0	0	0	1	0
0	0	1	0	0
0	0	1	1	1
0	1	0	0	0
0	1	0	1	0
0	1	1	0	0
0	1	1	1	1
1	0	0	0	0
1	0	0	1	0
1	0	1	0	0
1	0	1	1	1
1	1	0	0	1
1	1	0	1	1
1	1	1	0	1
1	1	1	1	1

- Light system around the parking is working using solar panels. To power on (store the energy) the solar panel there are 3 switches available. They are morning, daytime, and night-time. If you pressed the morning or the daytime switch the solar panel will work. But if you pressed the night switch the solar panel will stop working.

A = Morning, B = Daytime, C = Night-time

A	B	C	Z
0	0	0	0
0	0	1	0
0	1	0	1
0	1	1	0
1	0	0	1
1	0	1	0
1	1	0	1
1	1	1	0

• High-Security System

- When somebody needs to open the main gate, they will be scanned using a laser and they will be scanned using a laser and they must press a button to enter.

Scanner = A, Button = B

A	B	Z
0	0	0
0	1	0
1	0	0
1	1	1

- To turn on the electric fence there is a switch. Only officers and watchers can turn on that switch. Jailers can also turn on it, but watchers need to be around to do it.

Officer = A, Watcher = B, Jailer = C

A	B	C	Z
0	0	0	0
0	0	1	0
0	1	0	0
0	1	1	1
1	0	0	0
1	0	1	0
1	1	0	0
1	1	1	1

- There are 2 buttons to warn the officers that the electric fence is damaged. Both buttons must be pressed to indicate it.

Button1 = A, Button2 = B

A	B	Z
0	0	0
0	1	0
1	0	0
1	1	1

- To activate the roadblocks jailer must first press a button. Then he can activate it. Or, to activate it officer or a security guard can easily activate it without pressing any button.

Jailer = A, Button = B, Officer = C, Security Guard = D

A	B	C	D	Z
0	0	0	0	0
0	0	0	1	1
0	0	1	0	1
0	0	1	1	1
0	1	0	0	0
0	1	0	1	1
0	1	1	0	1
0	1	1	1	1
1	0	0	0	0
1	0	0	1	1
1	0	1	0	1
1	0	1	1	1
1	1	0	0	1
1	1	0	1	1
1	1	1	0	1
1	1	1	1	1

2.3 K maps

- Identify and count the prisoners

A\BC	00	01	11	10
0	0	0	0	1
1	1	0	0	1

$$\text{SOP Expression} = AC' + BC'$$

- Lighting System

- Turn on the lights, jailers, and officers' use switches. They both can turn it on and off anytime. Also, the security guard must turn on the lights at night.

A = Officer, B = Jailer, C = Security guard

A\BC	00	01	11	10
0	0	0	1	0
1	0	0	1	1

$$\text{SOP Expression} = A + B + C$$

- There are 3 buttons in the light system. 1st button presents the morning, 2nd button presents the daytime and the 3rd button presents the night-time. Only the lights will work if somebody presses the button 3.

Button1 = A, Button2 = B, Button3 = C

A\BC	00	01	11	10
0	0	1	1	0
1	0	1	0	0

$$\text{SOP Expression} = C(B' + A)$$

- The prison has an emergency button to indicate if an emergency occurred. At least two people need to press the button, and if you did it will turn on the lights around the prison. (People; watcher in the watch tower, officer, and the jailer)

A = Jailer, B = Officer, C = Watcher

A\BC	00	01	11	10
0	0	0	1	0
1	0	0	1	1

$$\text{SOP Expression} = B(A + C)$$

- **Water Tank System**

A\B	0	1
0	0	0
1	0	1

SOP Expression = AB

- **Security System**

- When both jailer and the officer turned on the emergency button, cells will be auto-locked, and a buzzer will work. They cannot activate it alone. But a watcher from a watch tower can.

A = Jailer, B = Officer, C = Watcher

A\BC	00	01	11	10
0	0	1	1	0
1	0	1	1	1

SOP Expression = $A + B + C$

- If the smoke detector and the heat detector both start to work or, if a worker pressed the switch (fire alarm bell switch), the alarm and the fire exit door will be opened.

A = Smoke Detector, B = Heat Detector, C = Switch Alarm

A\BC	00	01	11	10
0	0	1	1	0
1	0	1	1	1

SOP Expression = $A + B + C$

- **Vehicle Park System**

- 3 people are allowed to open the parking gate. They are security guards, jailers, and the officer. The jailer is not allowed to open the gate by himself, he needs to do it with the officer. The security guard can do it by himself.

A = Security guard, B = Jailer, C = Officer

A\BC	00	01	11	10
0	0	0	1	0
1	1	1	0	1

SOP Expression = $A'C + AC' + AB'$

- Add a password using 3 switches. If all the 3 switches were pressed, the parking gate will open. Also, the vehicle's number plate will be scanned too.

A = Switch1, B = Switch2, C = Switch3, D = Scanner

AB\CD	00	01	11	10
00	0	0	0	0
01	0	0	0	0
11	0	0	1	0
10	0	0	0	0

SOP Expression = **ABCD**

- To turn on the lights there are 4 switches. If somebody pressed switch 1 and switch 2, all the lights will be turned on. Also, switch 3 and 4 will turn on all the lights. If you pressed all the switches, then also the lights will turn on. Else lights will not turn on.

Switch1 = A, Switch2 = B, Switch3 = C, Switch4 = D

AB\CD	00	01	11	10
00	0	0	1	0
01	0	0	1	0
11	1	1	1	1
10	0	0	1	0

SOP Expression = **AB + CD**

- The light system around the parking is working using solar panels. To power on (store the energy) the solar panel there are 3 switches available. They are morning, daytime, and night-time. If you pressed the morning or the daytime switch the solar panel will work. But if you pressed the night switch the solar panel will stop working.

A = Morning, B = Daytime, C = Night-time

A\BC	00	01	11	10
0	0	1	1	1
1	0	0	0	0

SOP Expression = **BC'**

- **High-Security System**

- When somebody needs to open the main gate, they will be scanned using a laser and they will be scanned using a laser and they must press a button to enter.

Scanner = A, Button = B

A\B	0	1
0	0	0
1	0	1

SOP Expression = **AB**

- To turn on the electric fence there is a switch. Only officers and watchers can turn on that switch. Jailers can also turn on it, but watchers need to be around to do it.

Officer = A, Watcher = B, Jailer = C

A\BC	00	01	11	10
0	0	0	1	0
1	0	0	1	0

SOP Expression = **BC**

- There are 2 buttons to warn the officers that the electric fence is damaged. Both buttons must be pressed to indicate it.

Button1 = A, Button2 = B

A\B	0	1
0	0	0
1	0	1

SOP Expression = **AB**

- To activate the roadblocks jailer must first press a button. Then he can activate it. Or, to activate it officer or a security guard can easily activate it without pressing any button.

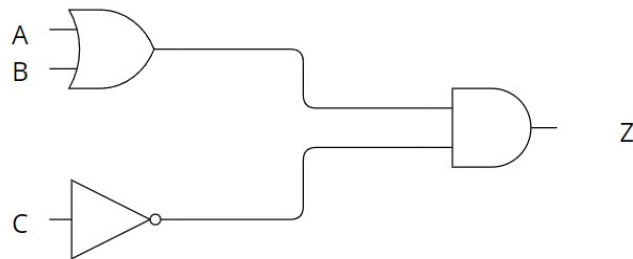
Jailer = A, Button = B, Officer = C, Security Guard = D

AB\CD	00	01	11	10
00	0	1	1	1
01	0	1	1	1
11	1	1	1	1
10	0	1	1	1

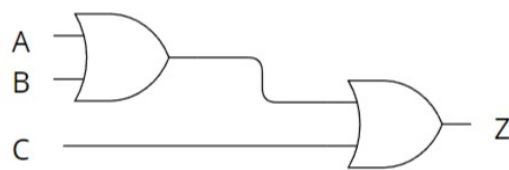
SOP Expression = **AB + C + D**

2.4 Circuit Diagrams

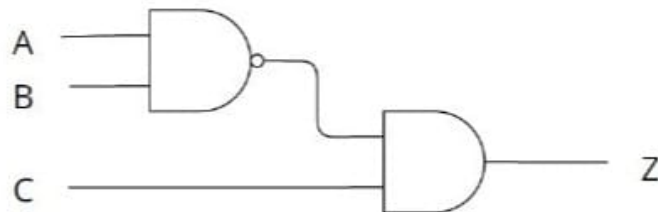
- **Identify and count the prisoners**



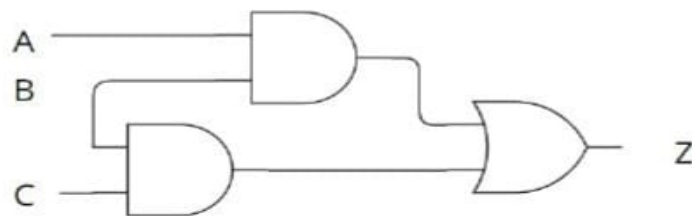
- **Lighting System**



Turn on the lights, jailers, and officers' use switches. They both can turn it on and off anytime. Also, the security guard must duty turn on the lights at night.

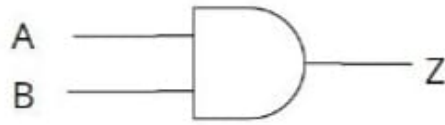


There are 3 buttons in the light system. 1st button presents the morning, 2nd button presents the daytime and the 3rd button presents the night-time. Only the lights will work if somebody presses the button 3.

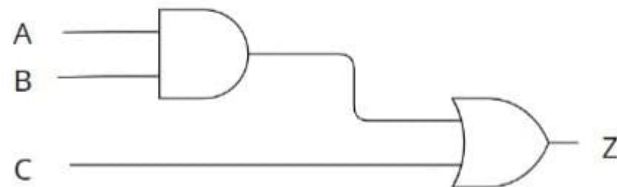


The prison has an emergency button to indicate if an emergency occurred. At least two people need to press the button, and if you did it will turn on the lights around the prison. (People; watchers in the watch tower, officer, and the jailer)

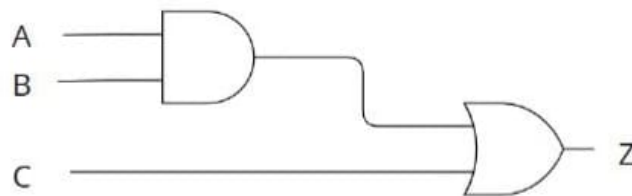
- **Water Tank System**



- **Security System**

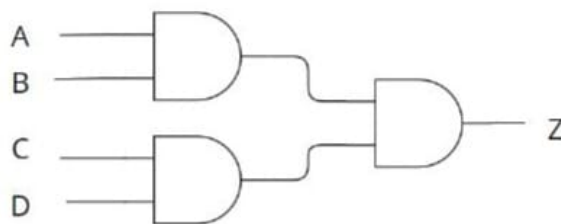


When both jailer and the officer turned on the emergency button, cells will be auto-locked, and a buzzer will work. They cannot activate it alone. But a watcher from a watch tower can.

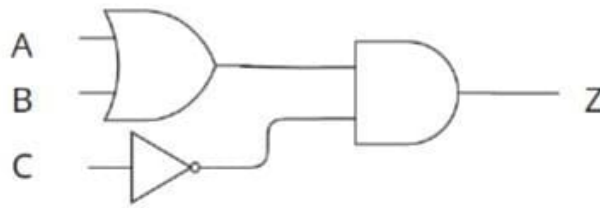


If the smoke detector and the heat detector both start to work or, if a worker pressed the switch (fire alarm bell switch), the alarm and the fire exit door will be opened.

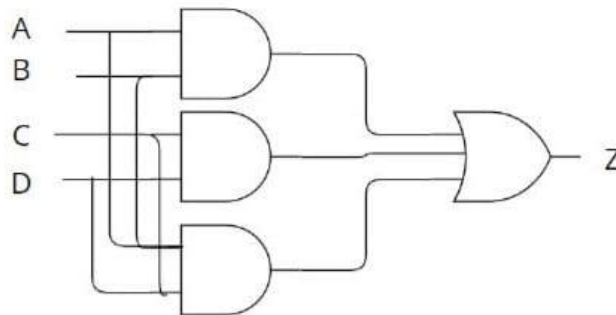
- **Vehicle Park System**



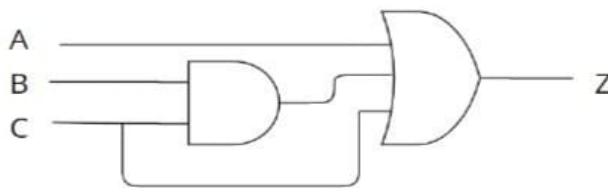
Add a password using 3 switches. If all the 3 switches were pressed, the parking gate will open. Also, the vehicle's number plate will be scanned too.



The light system around the parking is working using solar panels. To power on (store the energy) the solar panel there are 3 switches available. They are morning, daytime, and night-time. If you pressed the morning or the daytime switch the solar panel will work. But if you pressed the night switch the solar panel will stop working.

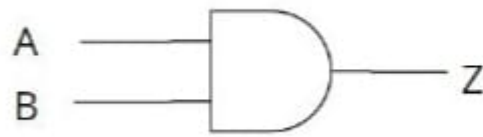


To turn on the lights there are 4 switches. If somebody pressed switch 1 and switch 2, all the lights will be turned on. Also, switch 3 and 4 will turn on all the lights. If you pressed all the switches, then also the lights will turn on. Else lights will not turn on.

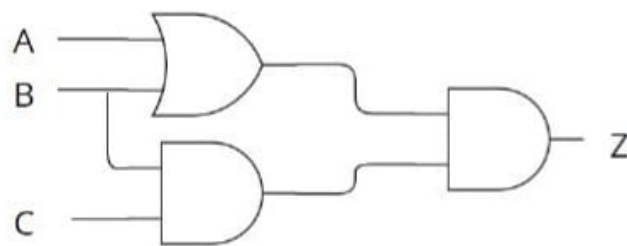


3 people are allowed to open the parking gate. They are security guards, jailers, and the officer. The jailer is not allowed to open the gate by himself, he needs to do it with the officer. The security guard can do it by himself.

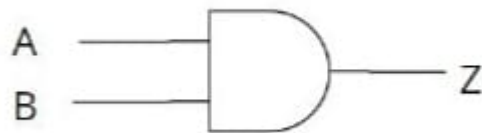
- **High-Security System**



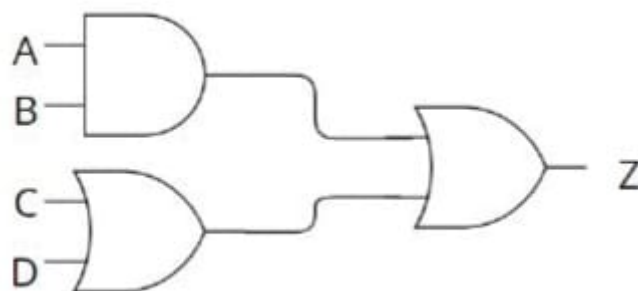
When somebody needs to open the main gate, they will be scanned using a laser and they will be scanned using a laser and they must press a button to enter.



To turn on the electric fence there is a switch. Only officers and watchers can turn on that switch. Jailers can also turn on it, but watchers need to be around to do it.



There are 2 buttons to warn the officers that the electric fence is damaged. Both buttons must be pressed to indicate it.



To activate the roadblocks jailer must first press a button. Then he can activate it. Or, to activate it officer or a security guard can easily activate it without pressing any button.

CHAPTER 03

3.1 Results of the operation

- In this prison system, we have focused on 3 prominent topics that play a key role in a prison system. They are security, sustainability, safety systems, and power management.
- The security system we have used a High-Security system using lasers that will immediately start an emergency alarm in case a prisoner tries to escape. added electric fences making them more secure.
- Under the security system, we have added functions to identify and count the prisoners in case a prisoner went missing after the working period. if the system notes an off prisoner the system will indicate by red led and a buzzer.
- We have added an emergency but to use in critical situations. cells will be auto-locked along with the gates outside making no one escape or enter. buzzers and LEDs will start working right after activation.
- Added a vehicle park system with a security system that has a password using three switches. The parking gate can be opened when all three switches are pressed at the same time. making the security even strong.
- Added roadblocks in case of critical situations. Activation starts right after pressing a button and only officers and jailors have access to this function.
- Underwater management, we have added a water tank that supplies water to the prison. When a worker turns on the switch, the water tank will start to work. The worker is only allowed to turn on the tank in the daytime.
- Under the power management system, we have added a lighting system inside the cells. Jailors and officers use switches to turn on and off lights anytime.
- Under sustainability, we have added solar panels that charge a battery that gives power to lights around the prison.

CHAPTER 04

4.1 Limitations, Recommendations, and Conclusions

- **Limitations**

Using Sensors

In this prison project, we have introduced some features that need to use sensors to detect motion, heat, smoke, scan, and darkness. But as our project, we are not allowed to use them. So, to represent the sensors, we used LED lights and Switches.

Power Supply

To indicate functions, we mostly use LED lights. So, we had to use a power supply separately for LEDs and Other components because LEDs can only consume power in the range of 2V-3.5V. if we gave high voltage the LEDs in the model can be damaged. So, to give the exact voltage we use an Arduino Uno Board. (5V)

- **Recommendations**

Use dedicated real Sensors

- ✓ Motion detects sensors to detect motions
- ✓ Smoke and heat detection sensors for the fire alarm system
- ✓ LDR sensors to detect darkness
- ✓ Scanners for scanning doors
- ✓ Lasers for the High-Security area

Use Coding for Arduino Uno Board

If we used coding for our Arduino Uno Board, we could add limitations and make it more difficult and advanced.

- **Conclusion**

We have designed this model very simply to represent all the functions using LEDs, buzzers, and switches. We have not used any sensors or proper power supply methods while building the model according to the limitations. When building this as a real-world project we recommend investing in highly advanced and genuine sensors, and power supply methods. and with the knowledge of software engineers build a proper program to interact with sensors and security features properly. In this way, we hope this prison system will be more accurate and advanced.

References

We completed our report using different sites on the internet. They are,

- ✓ [To get information about the prison system.](#)
- ✓ [To get an idea about circuit drawing](#)
- ✓ [To draw the circuit diagrams](#)